

Supplementary: Mathematical Proofs

Supplementary: Mathematical Proofs I

This supplementary material provides complete formal proofs for all theorems stated in the main text, including preliminary definitions (sec:preliminaries), main theorem proofs (sec:thm42-proofsec:thm511-proof), and additional supporting lemmas (sec:additional-lemmas).

Preliminary Definitions and Notation I

Preliminary Definitions and Notation II

Notation Summary

Table 1: Mathematical notation used throughout proofs.

Symbol	Meaning
$\mathcal{A} = \{a_1, \dots, a_n\}$	Set of n agents
$\mathcal{B}_i : \Phi \rightarrow [0, 1]$	Agent i 's belief function
$\mathcal{G}_i$	Agent i 's goal set
$\mathcal{T}_{i \rightarrow j}$	Trust from agent i to agent j
$\delta \in (0, 1)$	Trust decay factor per delegation hop
τ	Generic threshold parameter
ϕ, ψ	Propositions
$\pi(\phi)$	Provenance chain for belief ϕ

Definition (Trust Path)

A trust path from agent a_0 to agent a_k is an ordered sequence $p = (a_0, a_1, \dots, a_k)$ where each consecutive pair (a_i, a_{i+1}) represents a direct trust relationship with $\mathcal{T}_{a_i \rightarrow a_{i+1}} > 0$.

Definition (Path Trust)

Theorem 4.2: Trust Boundedness I

Theorem (Trust Boundedness — Restated)

For any delegation chain of depth d :

$$\mathcal{T}_{i \rightarrow k}^{del} \leq \delta^d \quad (2)$$

Lemma (Trust Non-Amplification on Single Hop)

For any agents a, b and any delegation to c :

$$\mathcal{T}_{a \rightarrow c}^{del} \leq \mathcal{T}_{a \rightarrow b} \quad (3)$$

Theorem 4.2: Trust Boundedness II

Proof of lem:single-hop.

By the trust delegation rule (def:trust-delegation):

$$\mathcal{T}_{a \rightarrow c}^{del} = \min(\mathcal{T}_{a \rightarrow b}, \mathcal{T}_{b \rightarrow c}) \cdot \delta \quad (4)$$

Since $\min(\mathcal{T}_{a \rightarrow b}, \mathcal{T}_{b \rightarrow c}) \leq \mathcal{T}_{a \rightarrow b}$ and $\delta < 1$:

$$\mathcal{T}_{a \rightarrow c}^{del} = \min(\mathcal{T}_{a \rightarrow b}, \mathcal{T}_{b \rightarrow c}) \cdot \delta \leq \mathcal{T}_{a \rightarrow b} \cdot \delta < \mathcal{T}_{a \rightarrow b} \quad (5)$$

□

Lemma (Trust Decay Bound)

For any single-hop delegation:

$$\mathcal{T}^{del} \leq \delta \quad (6)$$

Theorem 4.2: Trust Boundedness III

Proof of lem:decay-bound.

By definition, all direct trust values satisfy $\mathcal{T}_{a \rightarrow b} \leq 1$. Therefore:

$$\mathcal{T}_{a \rightarrow c}^{del} = \min(\mathcal{T}_{a \rightarrow b}, \mathcal{T}_{b \rightarrow c}) \cdot \delta \leq 1 \cdot \delta = \delta \quad (7)$$



Theorem 4.2: Trust Boundedness IV

Main Proof of thm:trust-bound-restated.

By strong induction on d .

Base Case ($d = 0$): When $d = 0$, there is no delegation (direct trust). By definition:

$$\mathcal{T}_{i \rightarrow k}^{del} = \mathcal{T}_{i \rightarrow k} \leq 1 = \delta^0 \quad (8)$$

The base case holds.

Inductive Hypothesis: Assume for all delegation chains of depth $\leq d$:

$$\mathcal{T}^{del} \leq \delta^d \quad (9)$$

Inductive Step (depth $d + 1$): Consider a delegation chain $(a_0, a_1, \dots, a_{d+1})$ of depth $d + 1$.

Let $\mathcal{T}^{(d)}$ denote the delegated trust from a_0 to a_d (depth d).

By the trust delegation rule:

$$\mathcal{T}_{a_0 \rightarrow a_{d+1}}^{del} = \min(\mathcal{T}^{(d)}, \mathcal{T}_{a_d \rightarrow a_{d+1}}) \cdot \delta \quad (10)$$

By the inductive hypothesis: $\mathcal{T}^{(d)} < \delta^d$

Theorem 5.7: Belief Injection Resistance I

Theorem (Belief Injection Resistance — Restated)

Under CIF with firewall detection rate r_f and sandboxing verification rate r_s :

$$P(\mathcal{A}_{BI} \text{ succeeds}) \leq (1 - r_f) \cdot (1 - r_s) \quad (15)$$

Lemma (Defense Independence)

The firewall and sandbox operate on independent decision criteria:

- ▶ *Firewall: Pattern matching and anomaly scoring on message content*
- ▶ *Sandbox: Provenance verification, consistency checking, and corroboration*

These mechanisms share no common features or state.

Theorem 5.7: Belief Injection Resistance II

Proof of lem:defense-independence.

By construction of the CIF architecture:

1. Firewall operates at input layer with feature set
 $F_{\text{firewall}} = \{\text{patterns}, \text{embeddings}, \text{anomaly_scores}\}$
2. Sandbox operates at belief layer with feature set
 $F_{\text{sandbox}} = \{\text{provenance}, \text{consistency}, \text{corroboration}\}$
3. $F_{\text{firewall}} \cap F_{\text{sandbox}} = \emptyset$

Therefore,

$P(\text{firewall detects} | \text{sandbox outcome}) = P(\text{firewall detects})$. The mechanisms are probabilistically independent. $\square$

Definition (Attack Success)

A belief injection attack $\mathcal{A}_{BI}$ succeeds if and only if:

1. The adversarial message m_{adv} is not rejected by the firewall,
AND
2. The injected belief ϕ_{adv} is promoted from sandbox to verified beliefs

Theorem 5.7: Belief Injection Resistance III

Main Proof of thm:belief-injection-restated.

Let E_f = event “firewall accepts message” (does not detect attack). Let E_s = event “sandbox fails to filter belief” (does not detect attack).

For $\mathcal{A}_{BI}$ to succeed, both E_f and E_s must occur:

$$P(\mathcal{A}_{BI} \text{ succeeds}) = P(E_f \cap E_s) \quad (16)$$

By lem:defense-independence (independence):

$$P(E_f \cap E_s) = P(E_f) \cdot P(E_s) \quad (17)$$

By definition of detection rates:

► $P(E_f) = 1 - r_f$ (probability firewall misses attack)

► $P(E_s) = 1 - r_s$ (probability sandbox misses attack)

Therefore:

$$P(\mathcal{A}_{BI} \text{ succeeds}) = (1 - r_f) \cdot (1 - r_s) \quad (18)$$



Theorem 4.7: No Trust Amplification I

Theorem (No Trust Amplification — Restated)

For any path $p = (a_0, a_1, \dots, a_k)$ in the communication graph:

$$\mathcal{T}_{a_0 \rightarrow a_k}^{path} \leq \min_{i \in [0, k-1]} \mathcal{T}_{a_i \rightarrow a_{i+1}} \quad (21)$$

Lemma (Minimum Preservation under Min)

For any sequence $(x_1, \dots, x_n)$ and additional element x_{n+1} :

$$\min(x_1, \dots, x_{n+1}) = \min(\min(x_1, \dots, x_n), x_{n+1}) \quad (22)$$

Proof.

Standard property of the minimum function.



Theorem 4.7: No Trust Amplification II

Lemma (Decay Factor Strengthens Bound)

For $x \leq y$ and $\delta \in (0, 1)$:

$$x \cdot \delta \leq y \tag{23}$$

Proof.

Since $\delta < 1$, $x \cdot \delta < x \leq y$.



Theorem 4.7: No Trust Amplification III

Main Proof of thm:trust-amp-restated.

By strong induction on path length k .

Base Case ($k = 1$): For path $p = (a_0, a_1)$:

$$\mathcal{T}_{a_0 \rightarrow a_1}^{path} = \mathcal{T}_{a_0 \rightarrow a_1} = \min_{i \in [0,0]} \mathcal{T}_{a_i \rightarrow a_{i+1}} \quad (24)$$

The base case holds trivially.

Inductive Hypothesis: Assume for all paths of length $\leq k$:

$$\mathcal{T}^{path} \leq \min_{i \in [0, k-1]} \mathcal{T}_{a_i \rightarrow a_{i+1}} \quad (25)$$

Inductive Step (path length $k + 1$): Consider path

$p = (a_0, a_1, \dots, a_{k+1})$.

Let $p' = (a_0, a_1, \dots, a_k)$ be the prefix path.

By the trust delegation rule:

$$\mathcal{T}_{a_0 \rightarrow a_{k+1}}^{path} = \min(\mathcal{T}_{a_0 \rightarrow a_k}^{path'}, \mathcal{T}_{a_k \rightarrow a_{k+1}}) \cdot \delta \quad (26)$$

By the inductive hypothesis:

Theorem 5.8: Goal Alignment Invariant I

Theorem (Goal Alignment Invariant — Restated)

If the system starts with aligned goals and all goal updates follow the delegation protocol:

$$\text{Aligned}(\mathcal{G}_i^0) \wedge \forall t : \text{ValidUpdate}(\mathcal{G}_i^t, \mathcal{G}_i^{t+1}) \Rightarrow \forall t : \text{Aligned}(\mathcal{G}_i^t) \quad (32)$$

Definition (Goal Alignment)

Goals $\mathcal{G}_i$ are aligned if:

$$\text{Aligned}(\mathcal{G}_i) \iff \mathcal{G}_i \subseteq \mathcal{G}_{\text{principal}} \cup \text{Delegate}(\mathcal{G}_{\text{principal}}) \quad (33)$$

Theorem 5.8: Goal Alignment Invariant II

Definition (Valid Goal Update)

An update from $\mathcal{G}^t$ to $\mathcal{G}^{t+1}$ is valid if:

$$\text{ValidUpdate}(\mathcal{G}^t, \mathcal{G}^{t+1}) \iff \forall g \in (\mathcal{G}^{t+1} \setminus \mathcal{G}^t) : \text{Authorized}(g) \quad (34)$$

where $\text{Authorized}(g)$ means g derives from principal or valid delegation.

Lemma (Delegation Preserves Alignment)

If $g \in \text{Delegate}(\mathcal{G}_{\text{principal}})$, then g is aligned.

Proof.

Direct from def:goal-alignment. □

Lemma (Set Union Preserves Subset)

If $A \subseteq C$ and $B \subseteq C$, then $A \cup B \subseteq C$.

Theorem 5.8: Goal Alignment Invariant III

Proof.

Standard set theory.



Theorem 5.8: Goal Alignment Invariant IV

Main Proof of thm:goal-alignment-restated.

By induction on time t .

Base Case ($t = 0$): Given: $\text{Aligned}(\mathcal{G}_i^0)$. The base case holds by hypothesis.

Inductive Hypothesis: Assume $\text{Aligned}(\mathcal{G}_i^t)$ for some $t \geq 0$.

Inductive Step: We must show $\text{Aligned}(\mathcal{G}_i^{t+1})$.

The goal set at $t + 1$ is:

$$\mathcal{G}_i^{t+1} = (\mathcal{G}_i^t \setminus \text{Removed}) \cup \text{Added} \quad (35)$$

For goals in $\mathcal{G}_i^t \setminus \text{Removed}$:

- ▶ By inductive hypothesis, these are aligned
- ▶ Removal cannot introduce misalignment

For goals in Added :

- ▶ By `ValidUpdate`, all added goals satisfy $\text{Authorized}(g)$
- ▶ By `lem:delegation-preserves`, authorized goals are aligned

By `lem:union-preserves`:

$$\mathcal{G}_i^{t+1} \subseteq \mathcal{G}_{\text{principal}} \cup \text{Delegate}(\mathcal{G}_{\text{principal}}) \quad (36)$$

Theorem 5.9: Firewall Liveness I

Theorem (Firewall Liveness — Restated)

CIF firewall preserves liveness for legitimate inputs:

$$\forall m \in \mathcal{M}_{\text{legitimate}} : P(\mathcal{F}(m) = \text{ACCEPT}) \geq 1 - \epsilon_{fp} \quad (38)$$

Definition (Legitimate Message)

A message m is legitimate if:

1. It originates from an authorized source
2. It contains no adversarial content
3. It conforms to expected communication patterns

Definition (False Positive Rate)

The false positive rate ϵ_{fp} is:

$$\epsilon_{fp} = P(\mathcal{F}(m) \neq \text{ACCEPT} | m \in \mathcal{M}_{\text{legitimate}}) \quad (39)$$

Theorem 5.9: Firewall Liveness II

Lemma (Firewall Classification)

For any message m , the firewall produces exactly one of three outcomes:

$$\mathcal{F}(m) \in \{ACCEPT, QUARANTINE, REJECT\} \quad (40)$$

Proof.

By construction of the firewall decision function (def:firewall). □

Theorem 5.9: Firewall Liveness III

Main Proof of thm:firewall-liveness-restated.

Let $m \in \mathcal{M}_{legitimate}$ be an arbitrary legitimate message.

By the law of total probability:

$$P(\mathcal{F}(m) = \text{ACCEPT}) + P(\mathcal{F}(m) = \text{QUARANTINE}) + P(\mathcal{F}(m) = \text{REJECT}) \quad (41)$$

By def:false-positive:

$$P(\mathcal{F}(m) \neq \text{ACCEPT}) = \epsilon_{fp} \quad (42)$$

Therefore:

$$P(\mathcal{F}(m) = \text{ACCEPT}) = 1 - P(\mathcal{F}(m) \neq \text{ACCEPT}) = 1 - \epsilon_{fp} \quad (43)$$

Since m was arbitrary:

$$\forall m \in \mathcal{M}_{legitimate} : P(\mathcal{F}(m) = \text{ACCEPT}) \geq 1 - \epsilon_{fp} \quad (44)$$



Corollary (Availability Bound)

Theorem 5.10: Byzantine Consensus Termination I

Theorem (Byzantine Consensus Termination — Restated)

With $n \geq 3f + 1$ agents and at most f Byzantine:

$$P(\text{consensus reached in } O(f + 1) \text{ rounds}) = 1 \quad (45)$$

Lemma (Byzantine Agreement Bound)

Byzantine agreement requires $n \geq 3f + 1$ to tolerate f Byzantine agents.

Proof.

Classical result from distributed systems (Lamport, Shostak, Pease 1982). With fewer agents, Byzantine agents can equivocate and prevent agreement. □

Theorem 5.10: Byzantine Consensus Termination II

Lemma (Honest Majority)

With $n \geq 3f + 1$:

$$n - f \geq 2f + 1 > \frac{2n}{3} \quad (46)$$

Proof.

$$n - f \geq (3f + 1) - f = 2f + 1$$

$$\frac{2n}{3} \leq \frac{2(3f+1)}{3} = 2f + \frac{2}{3} < 2f + 1$$

Therefore $n - f > \frac{2n}{3}$. □

Lemma (Round Progression)

In each round, at least one of the following occurs:

1. *Consensus is reached, or*
2. *At least one Byzantine agent is detected and excluded*

Theorem 5.10: Byzantine Consensus Termination III

Proof.

By the protocol structure:

- ▶ If honest agents agree, their majority ($> 2n/3$) ensures consensus
- ▶ If no consensus, some agent must have equivocated
- ▶ Equivocation is detectable through signature verification



Theorem 5.10: Byzantine Consensus Termination IV

Main Proof of thm:byzantine-restated.

Termination: By lem:round-progression, each round without consensus excludes at least one Byzantine agent.

With at most f Byzantine agents, at most f rounds can occur without consensus.

After f exclusions, all remaining agents are honest.

By lem:honest-majority, honest agents form a $> 2/3$ majority and reach consensus in one additional round.

Total rounds: at most $f + 1 = O(f + 1)$.

Probability: The protocol is deterministic given message delivery. With reliable (eventually synchronous) channels, all messages are delivered.

Therefore, termination is guaranteed with probability 1. □

Corollary (Concrete Round Bound)

With $f = 2$ Byzantine agents: consensus in at most 3 rounds.

Theorem 5.10: Byzantine Consensus Termination V

Corollary (Safety)

All honest agents decide on the same value (agreement property).

Proof.

By honest majority and the $2/3$ threshold requirement.



Theorem 5.11: Bounded Overhead I

Theorem (Bounded Overhead — Restated)

CIF adds latency:

$$L_{CIF} = L_{firewall} + L_{sandbox} \cdot P(\textit{quarantine}) + L_{verify} \cdot P(\textit{verify}) \quad (47)$$

Definition (Message Processing Path)

A message m follows one of three paths:

1. **Accept path:** Firewall check only
2. **Quarantine path:** Firewall + sandbox processing
3. **Reject path:** Firewall check only (early termination)

Theorem 5.11: Bounded Overhead II

Lemma (Expected Value Decomposition)

For mutually exclusive events E_1, E_2, E_3 with $\sum P(E_i) = 1$:

$$E[L] = \sum_i P(E_i) \cdot L_i \quad (48)$$

Proof.

Law of total expectation.



Theorem 5.11: Bounded Overhead III

Main Proof of thm:overhead-restated.

Let:

- ▶ L_{firewall} = firewall processing latency
- ▶ L_{sandbox} = sandbox processing latency
- ▶ L_{verify} = provenance verification latency
- ▶ $P_q = P(\text{quarantine})$ = probability of quarantine
- ▶ $P_v = P(\text{verify})$ = probability verification is triggered

The total CIF latency is:

$$L_{CIF} = L_{\text{firewall}} + \mathbb{1}[\text{quarantine}] \cdot L_{\text{sandbox}} + \mathbb{1}[\text{verify}] \cdot L_{\text{verify}} \quad (49)$$

Taking expectations:

$$\begin{aligned} E[L_{CIF}] &= E[L_{\text{firewall}}] + E[\mathbb{1}[\text{quarantine}]] \cdot L_{\text{sandbox}} + E[\mathbb{1}[\text{verify}]] \cdot L_{\text{verify}} \\ &= L_{\text{firewall}} + P_q \cdot L_{\text{sandbox}} + P_v \cdot L_{\text{verify}} \end{aligned} \quad (50)$$



Numerical Instantiation

Additional Lemmas I

Lemma (Provenance Chain Integrity)

If provenance verification function V is a cryptographic hash chain, then:

$$V(\pi(\phi)) = 1 \Rightarrow \pi(\phi) \text{ has not been tampered with} \quad (54)$$

Proof.

By properties of cryptographic hash functions:

1. Collision resistance: Cannot find $\pi' \neq \pi$ with $H(\pi') = H(\pi)$
2. Preimage resistance: Cannot construct valid π without knowledge of chain

Therefore, $V(\pi(\phi)) = 1$ implies $\pi(\phi)$ is the original, untampered chain. □

Additional Lemmas II

Lemma (Belief Consistency Decidability)

*For finite proposition set Φ and belief function $\mathcal{B} : \Phi \rightarrow [0, 1]$:
Checking $\text{Consistent}(\mathcal{B})$ is decidable in $O(|\Phi|^2)$.*

Proof.

For each pair $(\phi, \psi) \in \Phi \times \Phi$:

1. Check if $\phi \wedge \psi \vdash \perp$ (logical contradiction)
2. Check if both $\mathcal{B}(\phi) > \tau$ and $\mathcal{B}(\psi) > \tau$

There are $O(|\Phi|^2)$ pairs. Each check is $O(1)$ with precomputed contradiction table.

Total: $O(|\Phi|^2)$. □

Lemma (Trust Matrix Convergence)

Under stable interaction patterns, the reputation component T_{rep} converges:

$$\lim_{t \rightarrow \infty} T_{rep}^t = T_{rep}^* \quad (55)$$

where T_{rep}^ reflects the agent's true reliability.*

Additional Lemmas III

Proof.

The reputation update rule is:

$$T_{rep}^{t+1} = T_{rep}^t + \eta \cdot (\text{outcome}_t - T_{rep}^t) \quad (56)$$

This is an exponential moving average with learning rate η .
For i.i.d. outcomes with mean μ :

$$E[T_{rep}^t] \rightarrow \mu \text{ as } t \rightarrow \infty \quad (57)$$

By the strong law of large numbers, $T_{rep}^t \rightarrow \mu$ almost surely. □

Summary of Proof Techniques I

Table 2: Summary of proof techniques by theorem.

Theorem	Primary Technique	Complexity
3.1 (Trust Boundedness)	Strong induction	$O(d)$
6.1 (Belief Injection Resistance)	Probability independence	$O(1)$
6.2 (No Trust Amplification)	Strong induction	$O(k)$
6.3 (Goal Alignment Invariant)	Induction on time	$O(t)$
6.4 (Firewall Liveness)	Complement probability	$O(1)$
6.5 (Byzantine Consensus)	Classical BFT	$O(f)$
6.6 (Bounded Overhead)	Expected value	$O(1)$

All proofs are constructive and provide explicit bounds useful for system implementation and analysis.